[Research Papers -Unpublished]

AI-Driven Digital Product Innovation Practice for Specific Population Groups in Community Healthcare Settings — A Case Study on Creating Medical Maternal and Child Care Tools

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Status: Unpublished Working Paper
Context: Independent research at Alibaba Health

Abstract: This research focuses on innovative design of intelligent digital products in healthcare, specifically examining AI-driven digital product innovation toolsets for maternal and child care groups. Through in-depth user research, innovative design strategies, and practical case analysis, we explore how to meet the diverse needs of maternal and child groups, enhance user experience, and strengthen product competitiveness and impact. The research covers inclusive design, AI intelligent design, and therapeutic emotional design in healthcare. Results indicate that user-centered design principles, intelligent technology applications, scenario-based service models, and emotional design elements are key factors in achieving innovative design for maternal and child care tools. These elements effectively address pain points in family childcare, such as family medication management and intelligent maternal-child dietary management, providing practical evidence for innovative design targeting specific populations in healthcare while supporting the development of maternal and child health services.

Keywords: Community Healthcare; Maternal and Child Health; AI Intelligence; Digital Products; Innovative Design

I. Introduction

(I) Research Background

Against the backdrop of rising global healthcare awareness, smart healthcare, digital medicine, and the integration of internet with healthcare have become mainstream trends. The key challenge lies in addressing various pain points faced by specific populations in family and community healthcare settings, thereby improving the digital healthcare experience and service quality for diverse user groups. In inclusive healthcare, maternal and child groups, elderly groups, and chronic disease management groups receive widespread social attention. With advancing intelligent technology and increasing popularity of internet healthcare, intelligent digital products are becoming widely applied in medical and healthcare fields, providing more convenient, efficient, and professional health services for maternal and child groups. As a special and substantial user group, maternal and child populations have unique and urgent healthcare service needs, opening up broad opportunities for innovative design of intelligent digital products in this field. From the user profile of post-90s new maternal groups, they demonstrate characteristics of intelligence, digitalization, and customization, pursuing precise services, efficient experiences, and refined parenting models.

(II) Research Purpose and Significance

This research focuses on innovative design of intelligent digital products in healthcare, using Alibaba Health's medical community maternal and child care tools as the research subject. Through practical case studies, it explores how to use digital innovation design to meet the diverse needs of maternal and child groups, enhance user experience, and strengthen product competitiveness and impact. This research will positively promote community healthcare scenarios for maternal-child health care and family health management, while opening new perspectives and approaches for design disciplines in healthcare direction, promoting cross-disciplinary development. The research emphasizes exploring intelligent design, emotional healing, and inclusive design for specific populations in healthcare.

II. Project Overview

(I) Project Background

This project is dedicated to developing professional and reliable family healthcare intelligent tools and feeding products for maternal and child users aged 0-3 years. The project aims to build a professional user ecosystem, achieving win-win business and user value through enhanced user engagement and loyalty. The project team conducted detailed research on maternal and child users' needs, pain points, and behavioral characteristics, developing a series of innovative maternal and child tools to meet various needs of this special group in healthcare.

Alibaba Health's medical community digital products provide online consultation, appointment booking, vaccine and health check scheduling, and medicine delivery services, while integrating related resources to provide medical content search and internet healthcare services. Business data shows significant growth in children's vaccination business with continued promotion. Based on vaccine services, offline community mother groups continue to expand, with users through community hospitals accounting for 57% of monthly active users (MAU). The core user group is mothers, with women accounting for 59%, and women aged 19-35 accounting for 45%. These users show high engagement, with average monthly active days reaching 3.4. The maternal and child population (new mothers) shows four significant characteristics: large scale, strong stickiness, predominantly new generation, and high population focus. We will conduct further research on maternal and child groups, exploring more opportunities to incubate innovative digital products and service experiences for maternal and child groups in medical community scenarios.

(II) Project Objectives

The business objective for the medical health community's maternal and child group is to design consumption paths based on mothers' special needs at different life cycle stages and continuously iterate related products. The maternal and child tool products' business objective is to meticulously plan tool-type products for the medical community platform, effectively connecting and penetrating maternal and child users, focusing on meeting high-frequency and rigid demands of mothers with children aged 0-7 years. The project planning advances in three phases: First, helping new mothers efficiently solve common issues within the community; Second, providing 1-2 high-stickiness service items ensuring 70% of mothers continue using related services; Third, deepening service experience to provide comprehensive care for mothers of all age groups. Below are the specific project objectives.

1. Deeply understand maternal and child user groups' needs, pain points, and behavioral characteristics to establish a solid foundation for innovative design.

2. Design innovative, practical, and attractive intelligent maternal and child digital tool products based on user data analysis and market trends, while creating matching quality service experiences.
3. Continuously improve products and enhance user experience through systematic user research, prototype design, testing, and ongoing iteration.
4. Explore innovative models and application scenarios for intelligent digital products in healthcare, providing reference for industry development.

III. Theoretical Foundation

(I) User-Centered Design Theory

In design, User-Centered Design (UCD) emphasizes comprehensive consideration of users' needs, capabilities, and usage contexts throughout the entire design process. This methodology provides a solid theoretical foundation for innovative design of maternal and child tools. Through in-depth research of maternal and child user groups, analyzing their behavioral characteristics, psychological states, and specific needs, designers can accurately grasp users' habits in different scenarios while clearly understanding their pain points. Based on these insights, designers can formulate reasonable design strategies ensuring final products effectively meet users' practical needs.

(II) Intelligent Design Theory

Intelligent design theory studies how to integrate advanced technologies like artificial intelligence and big data with product design to create product functions and service experiences with intelligent characteristics. For maternal and child tool design, intelligent design is mainly reflected in the following aspects:

1. Intelligent recommendation systems utilize big data analysis and machine learning algorithms to provide comprehensive and personalized recommendation services, such as recommending nutrition recipes and parenting knowledge based on baby's age and health conditions.
2. Leveraging AI's analytical capabilities to achieve intelligent decision support, providing scientific and reasonable decision support for maternal and child care. For example, providing preliminary diagnostic opinions based on baby's symptoms to help mothers make informed medical decisions.
3. For intelligent reminders and predictions, utilizing AI's time series analysis and prediction capabilities to provide convenient reminder services for maternal and child care, covering situations like specific vaccination times and accurate health check dates.

(III) Emotional Design Theory

Emotional design theory emphasizes that product design should not only meet users' functional needs but also focus on emotional experience. Through design methods to convey emotional value, it strengthens emotional connections between users and products. In maternal and child product design, emotional design is particularly important. As maternal and child groups often experience anxiety and fatigue during childcare, they especially need emotional support and detailed care. Through flexible application of emotional design and inclusive design concepts, we can provide

emotional healing experiences for maternal and child groups, help alleviate psychological pressure, while enhancing product recognition and stickiness in healthcare scenarios. This design concept enables us to better understand and respond to users' mental models at different stages and scenarios.

IV. User Research Methods

(I) User Research Methods

1. **Questionnaire Survey**: Design questionnaires for maternal and child user groups to collect basic information, usage habits, and needs and expectations for maternal and child tools. Questionnaires cover parenting experience, commonly used maternal and child products, function satisfaction, and issues during use. Through extensive online questionnaire research, obtain user data and behavioral characteristics to establish foundations for user profile construction and needs analysis.
2. **In-depth Interviews and Field Research**: Leveraging enterprise partnerships with community hospitals nationwide, select typical maternal and child users for in-depth interviews to understand their parenting experiences, pain points, and needs. Through interviews, guide users to share specific challenges in parenting process, discuss advantages and disadvantages of existing digital products, and expectations for ideal maternal and child tools. Observe user behavior in the field to grasp their emotional states and genuine thoughts. This in-depth research not only enriches data dimensions but also helps discover detailed issues that questionnaires might miss.
3. **User Profile Construction**: Based on questionnaire survey and in-depth interview data, construct maternal and child user profiles and user journey maps. User profiles include basic information (age, gender, region, family status, parenting stage), as well as parenting philosophy, consumption habits, and tool requirement characteristics. Through constructing typical profiles like "new mothers," "working mothers," "full-time mothers," design teams can intuitively understand target user group characteristics and needs differences, providing guidance for subsequent targeted design.
4. **Service Journey and Experience Mapping**: Analyze users' emotional experiences and pain points at different stages to discover needs and opportunities. Main aspects include: Usage scenario analysis: Record user behavior in different product usage scenarios, including the entire process from product discovery to long-term use; Emotional experience tracking: Track users' emotional changes at various stages to identify key factors affecting experience; Opportunity point mining: Identify product deficiencies and improvement opportunities for targeted product design optimization.

V. Maternal and Child Tool Design Strategies and Objectives

Through preliminary background research and user insights, we understand the basic characteristics and needs of maternal and child groups in the medical community. These maternal and child groups mainly consist of young mothers born in the 1985-1995 period. As a new generation of knowledge-based mothers, they generally trust authority, advocate scientific rationality, while emphasizing personal expression and thinking independently. In parenting, they display distinct characteristics: advocating refined parenting, emphasizing whole-family participation, preferring personalized feeding patterns, and pursuing diverse parenting philosophies and high-quality parenting products.

Therefore, creating a trustworthy and emotionally resonant exclusive maternal and child tool has become our core design objective.

(I) Building a Trustworthy Maternal and Child Tool

1. **Expert Endorsement**: Partner with authoritative medical institutions, invite pediatricians and nutritionists to participate in product development and content co-creation, establishing a scientifically rigorous brand image. Display medical qualification certification badges on the product interface to ensure credibility of medical advice; collaborate with universities and research institutions to publish parenting guidelines, strengthening content authority; create an "Online Experts" section, using real-time human interaction to address user concerns and bridge the professional gap.
2. **Content Trust Building**: Emphasize content professionalism and uniqueness, clearly indicate information sources such as references. Build user trust in the platform by providing accurate, scientific, and detailed content. For example, cite authoritative medical research or professional institutional guidelines in parenting articles to enhance content credibility.
3. **Brand Security Creation**: Strengthen user trust in the platform through the dual advantages of brand recognition and platform expertise. Specific measures include displaying brand certificates, showcasing quality partnership resources, and highlighting user testimonials to comprehensively enhance brand credibility.

(II) Achieving Efficient and Convenient Service Experience

1. **Intelligent Recommendation System**: Based on artificial intelligence and big data analysis algorithms, provide intuitive, personalized recommendation services. Analyze user browsing history, saved content, and other behavioral data to predict user interests and needs, precisely recommending relevant parenting knowledge, products, and services to improve information acquisition efficiency.

2. **Scenario-based Service Design**: Design various scenario-based service models around maternal and child users' different needs in various life scenarios. For example, provide medication guidance and nutritional meal suggestions in home parenting scenarios; offer vaccine appointment reminders and medication queries in outdoor scenarios. Through scenario-based services, better meet users' practical needs and provide more considerate, practical product experiences.

3. **Information Efficiency and Visualization**: Adopt simple and easy-to-use interaction methods, combining diverse presentation forms and design carriers such as graphics, videos, and charts to make content more vivid, intuitive, and easy to understand. Meanwhile, focus on information hierarchy and visual display to help users quickly access key information, improving information transmission efficiency and accuracy.

(III) Building Emotional and Caring Design

1. **Emotional Interface Design**: Use warm, soft color combinations and cute, approachable graphic elements and illustrations to create a comfortable, reassuring visual atmosphere that matches maternal and child products' positioning and users' psychological needs. Use warm color tones like light pink and light blue, paired with clean and clear interface layouts, to bring users a pleasant visual experience.

2. **Personalized Customization Services**: Provide personalized recommendation services and exclusive customization features based on users' personal preferences, behavioral habits, and needs. Allow users to customize theme colors, font sizes, and functional module layouts to meet different users' personalization needs and enhance product recognition and stickiness.
3. **Warm Interactive Experience**: Design thoughtful interaction details such as friendly language prompts, personalized greetings, and encouraging feedback messages to make users feel cared for and supported during use. Meanwhile, optimize interaction processes, reduce unnecessary steps and procedures, improve operation fluidity and efficiency, providing users with a convenient and comfortable interaction experience.

X. Conclusions and Future Outlook

(I) Research Conclusions

Through user research and practical exploration, this study developed an innovative community healthcare maternal and childcare tool product. This product met the healthcare needs of maternal and childcare users, resolved family feeding pain points, and enhanced user experience and product competitiveness. The research shows that the key to innovative design of intelligent digital products in healthcare lies in: user-centered design concepts, intelligent technology applications, scenario-based service models, and emotional design elements. These elements effectively solved problems faced by maternal and childcare users during parenting, providing a digital innovation case study for maternal and childcare groups in the vertical healthcare field.

(II) Future Outlook

With technological advancement and evolving user needs, intelligent digital products have broad prospects in healthcare. Future development will deepen inclusive and emotional design, focusing on research into diverse population needs including maternal and childcare, chronic disease management, elderly care, and accessibility services. Through exploring innovative design concepts and technology applications, we will improve product functions and service models to provide high-quality, efficient healthcare services for various population groups. Meanwhile, we will strengthen cooperation and exchange with medical institutions and experts, promote intelligent digital product innovation development, and contribute to healthcare advancement.

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